# ALTAF SHAIKH

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## Summary

Final Year B.Tech Computer Science & Engineering student and **Software Engineer** with hands-on experience designing, developing, and deploying full-stack applications using JavaScript, TypeScript, React.js, Node.js, Express.js, MongoDB, and Python. Experienced in building **REST APIs, real-time collaborative systems, authentication workflows, database-driven applications, and AI-powered software features**. Strong foundation in **Data Structures & Algorithms, Object-Oriented Programming, DBMS, and Operating Systems**, with experience solving engineering problems, debugging applications, and working with Git-based development workflows. Smart India Hackathon (SIH) 2025 Finalist and Google Student Ambassador with a strong interest in building reliable, scalable, and user-focused software products.

## Technical Skills

**Frontend Development:** React.js, JavaScript, TypeScript, HTML5, CSS3, Responsive Web Design, Tailwind CSS, Bootstrap
**Backend & APIs:** Node.js, Express.js, REST APIs, JWT Authentication
**Databases:** MongoDB
**Programming Languages:** JavaScript, TypeScript, Python, Java
**AI & Machine Learning:** LangChain, OpenAI API, Hugging Face, Sentence Transformers, FAISS, Ollama, NumPy, Pandas, Scikit-learn (Basics)
**Real-Time & Collaboration:** Socket.IO, Yjs, WebRTC
**Tools & Design:** Git, GitHub
**Core Computer Science:** Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Operating Systems

## Projects

• **SatQuery AI — Agentic Vision-Language Assistant** *GitHub*
*Python, LangChain, OpenAI-compatible APIs, Hugging Face, FAISS, Sentence Transformers, Ollama, pytest, Ruff*

- Built a research-oriented **agentic AI framework** that connects natural-language remote-sensing queries with retrieval, reasoning, and extensible specialist-tool workflows through an **agent controller**.
- Implemented modular **agent orchestration, knowledge-space and solution-space retrieval, toolkit registry, and configurable LLM integration** for extensible remote-sensing workflows.
- Developed retrieval components using **Sentence Transformers, embeddings, FAISS vector search, and DualRAG** for knowledge and solution retrieval.
- Added configurable support for **OpenAI APIs, Hugging Face, LangChain, and Ollama**, with automated testing through **pytest** and code-quality checks using **Ruff**.

• **SyncSpace — AI-Powered Collaborative Development Platform** *GitHub*
*React, Vite, Node.js, Express.js, MongoDB, Socket.IO, Yjs, WebRTC, OpenAI API*

- Architected a full-stack collaborative workspace combining **real-time coding, team communication, video meetings, whiteboarding, file sharing, and AI-assisted development**.
- Built a modular Node.js/Express backend with versioned **REST APIs**, authentication, room management, document handling, file management, code execution, replay, and analytics services.
- Implemented **AI Code Reviewer** and **AI Session Summary** workflows using OpenAI integration with local fallback behavior for supported functionality.
- Integrated **Socket.IO, Yjs, WebRTC, React Konva, and MongoDB** for real-time collaboration, synchronized editing, peer-to-peer communication, session replay, and analytics.

• **Netra AI — Assistive Vision & Obstacle Detection** *GitHub*
*Kotlin, Jetpack Compose, CameraX, MediaPipe, EfficientDet-Lite2, ML Kit*

- Built an Android assistive vision app that detects nearby obstacles through the phone camera and provides voice alerts to support visually impaired users.
- Implemented real-time object detection with MediaPipe and EfficientDet-Lite2, using temporal tracking, motion and blur filtering, and distance estimation to improve obstacle detection.
- Integrated English/Hindi text-to-speech, on-demand text recognition using ML Kit OCR, haptic alerts, and automatic flashlight control for hands-free assistance.

## Education

**Sanjeevan Group of Institutions** **2023 – 2027 (Expected)**
*B.Tech in Computer Science & Engineering — Final Year* *India*

- **Relevant Coursework:** Data Structures & Algorithms, DBMS, Operating Systems, OOP

**SAM Urdu High School** **2022**
*Class XII — Physics, Chemistry, Mathematics* *India*

## Achievements

- **Smart India Hackathon (SIH) 2025** — National-Level Finalist
- **BIG-O Battle: Vibe Coding SaaS Startup Challenge**, GDG VIT Mumbai — Top 2 Position
- **Buildverse Hackathon 2025**, HackWithIndia & GeeksforGeeks — Top 5 Position
- Participated in Google Cloud Agentic AI Day (powered by Hack2skill)

## Leadership & Activities

- **Google Student Ambassador (Team Gemini)**, Google — May 2026 – Present
- **Event Coordinator (Tech Management — Parikrama)**, Sanjeevan Group of Institutions — March 2026